

Industrials Practice

Turning humanoid supply chain constraints into billion-dollar wins

The robotics supply chain is the most underappreciated constraint on humanoid scale. But for companies that move now, it is the most significant opportunity in a generation.

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Humanoid robots are approaching an inflection point, with advances in AI, hardware, and investment accelerating their path from prototypes to real-world deployment. Venture capital funding for robotics surged more than threefold between 2023 and 2025, when it reached \$40.7 billion annually.¹ Governments have declared embodied AI a strategic priority, with China alone committing a \$138 billion state venture capital guidance fund to AI and robotics, among other high-tech sectors.² Both start-ups and established OEMs across Asia, Europe, and the United States are now scaling pilots in manufacturing, logistics, and industrial environments while training foundation models on real-world interaction data.

The strategic question is no longer whether humanoid robots will work but whether they can scale economically—and at what cost, speed, and reliability. In “[Humanoid robots: Crossing the chasm from concept to commercial reality](#),” we identify four bridges to commercial viability: radical cost reduction, greater dexterity and mobility, sustained uptime, and safety for fenceless operation. All four—cost reduction most directly—depend on a factor that receives less attention than AI capabilities: the maturity and scale of the underlying component supply chain.

For many of the most critical and costly humanoid components, the supplier ecosystem is still at the early stage for large-scale production, creating a significant opportunity for scale build-out. But as demand for humanoid hardware rises over the next few years, both suppliers and OEMs will have a greater incentive to increase investments. How the supply chain matures will shape the pace and economics of humanoid deployment over the next decade: where bottlenecks and opportunities exist, how sourcing strategies evolve, and which players will build the platforms the industry needs.

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¹ “State of venture 2025,” CBInsights, January 8, 2026.

² “China announces high-tech fund to grow AI, emerging industries,” CNN Newsource, March 6, 2025.

The anatomy of a humanoid robot's bill of materials

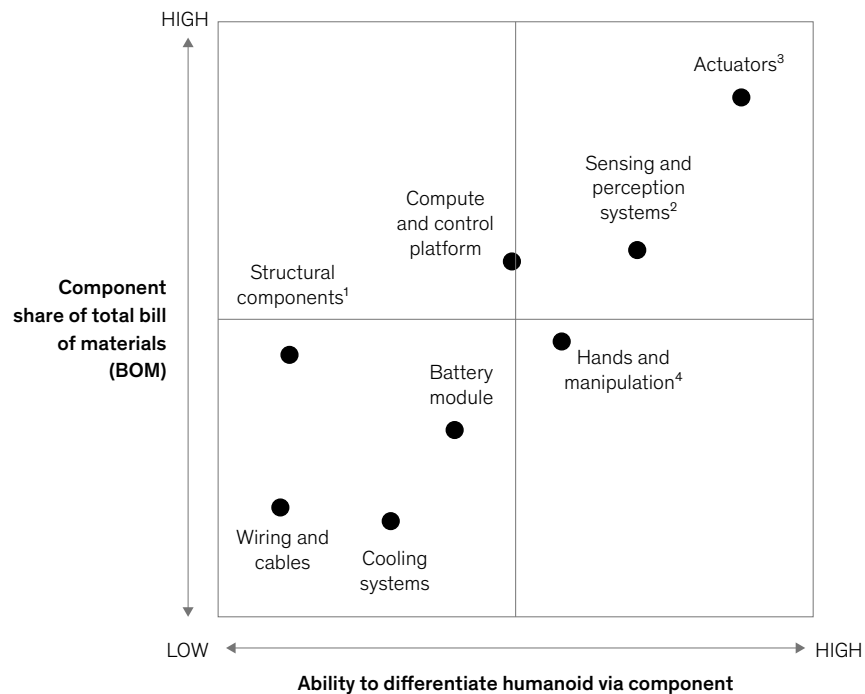
The core hardware stack of a humanoid spans five domains: actuators (40 to 60 percent of the bill of materials, or BOM), sensing and perception systems (10 to 20 percent), compute and control platforms (10 to 15 percent), structural components (5 to 10 percent), and battery modules (5 to 10 percent). Together, the five domains represent 85 to 90 percent of total unit costs, with the remainder coming from other areas, such as cooling systems and wiring harnesses (Exhibit 1). The impact of each domain on humanoid performance also varies.

Note that the “hands and manipulation” category in the chart includes only components that are procured separately from all other humanoid elements. Typically, hands are included in the sensor and actuation category. We show hands and manipulation here as a separate category

Exhibit 1

Five hardware domains are primarily responsible for humanoid costs and performance differentiation.

Component cost by level of component differentiation



Note: BOM impact of safety system is accounted for across various components, eg, safety-certified actuators and additional sensors/perception systems.

¹BOM impact depends on material (eg, composite vs alloy) and production method (eg, die casting vs forging); also includes plastic coverages.

²Cost differs significantly depending on component quality and component integration level.

³Structural/primary actuators.

⁴Only if procured separately; otherwise, hands and manipulation are part of actuators and sensors.

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Why hands are critical

Despite rapid progress in mobility and perception, humanoid hands remain one of the most challenging elements. While locomotion can be adapted to structured environments, most economically relevant tasks for hands, such as picking, sorting, or tool use, require reliable manipulation of diverse, unstructured objects.

Human hands combine more than 20 degrees of freedom with highly sensitive tactile feedback. Replicating this often requires dozens of miniaturized actuators—for example, Tesla's Optimus integrates more than 50 actuators in its hands alone, compared with roughly 25 to 30 across the rest of the body—alongside high-resolution force and tactile sensing and low-latency control, even though space and power are severely constrained.

From a supply chain perspective, hands sit at the intersection of multiple high-risk component categories. They depend on compact actuation and robotics-grade sensing, both of which still lack standardized architectures and fully scaled supplier ecosystems. This is particularly true for tactile sensing, where no dominant design has yet emerged. As a result, many developers rely on vertically integrated or highly customized designs.

Rather than a separate cost category, hands illustrate how bottlenecks emerge where performance requirements, supplier immaturity, and lack of standardization overlap. These factors make them a potential gating factor but also a key opportunity area when scaling humanoid deployment.

to demonstrate the importance of this subsystem as a stand-alone entity (see sidebar “Why hands are critical”).

What makes the supply chain picture distinctive is the mismatch between where value concentrates and where the supplier ecosystem is ready for high-volume production. Actuators, by far the largest cost driver and primary performance differentiator, depend on one of the least developed supplier ecosystems. Sensing, particularly force and tactile, faces similar exposure. By contrast, compute and battery components benefit from deep adjacency to the [electric vehicle \(EV\)](#), semiconductor, and consumer electronics industries, where production infrastructure already operates at scale.

Many high-impact humanoid components, including motors, actuator power electronics, permanent magnets, and precision bearings, overlap structurally with the EV value chain. This adjacency matters because it directly influences how quickly cost curves can compress as humanoid volumes increase. Subsystems with strong EV spillover benefit from existing tooling, supplier depth, and process maturity. Components without such adjacency, particularly robotics-grade force and tactile sensing, lack equivalent scale anchors and are more likely to face structural supply constraints.

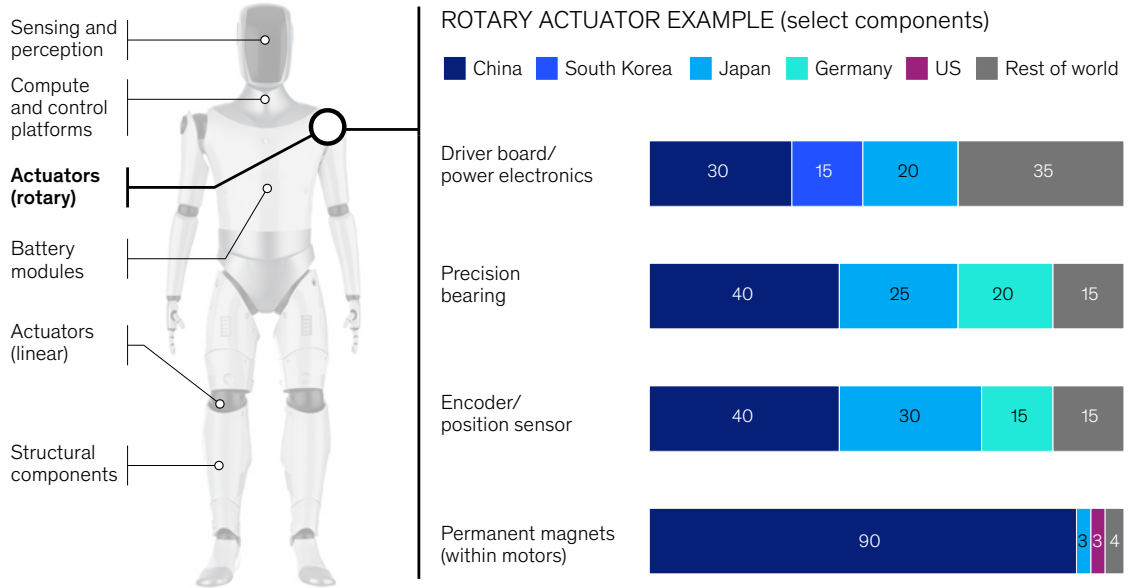
China, with its strong EV ecosystem, particularly benefits from the structural overlap and holds a significant share of capacity for several components used in humanoids (Exhibit 2). For more information on China's ecosystem advantage, see sidebar “China's distinctive supply chain model” on page 7.

Yet even where the underlying technology is well established in other industries, the humanoid supplier ecosystem is not ready for large-scale production. At present, suppliers have little incentive to increase production of humanoid-specific components because demand is too low to justify standardized, high-throughput manufacturing. This situation reinforces a scaling dilemma: Low volumes prevent suppliers from investing in dedicated production lines, yet without cost reductions, end user demand remains constrained. The typical humanoid BOM currently ranges from roughly \$30,000 to \$150,000 per unit. Costs under \$20,000 are widely cited as the long-term target, implying that significant cost compression is still required to unlock mass-market demand.

Exhibit 2

China has a significant share of capacity for robotics components.

Share of capacity for robotics component, by region, %



Note: Market share estimates are directional and triangulated across multiple third-party industry sources for all components.
Sources: McKinsey Battery Insights; McKinsey Center for Future Mobility; McKinsey Global Energy Insights

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In this premodular phase, most OEMs are forced into one of two paths to obtain components: vertical integration or close codevelopment with a limited number of partners in which they piece together adjacent industrial components and redesign them for humanoid requirements. Both paths accelerate development, but neither automatically translates into cost-efficient scale. Vertical integration concentrates all manufacturing overhead on a single OEM, while ad hoc codevelopment requires costly redesign that must be repeated for each platform.

Mapping supply risks and opportunities across the hardware stack

As humanoid deployments scale toward 2035, supply constraints and opportunities will not appear uniformly across the hardware stack. Instead, bottlenecks will concentrate in subsystems where high-performance requirements, supplier concentration, and limited standardization intersect and reinforce one another.

Understanding where constraints cluster and where the supplier base is already equipped for scale is essential for OEMs managing sourcing risk, suppliers prioritizing investment, and end users assessing deployment timelines. A McKinsey analysis of the current supply chain suggests that bottlenecks are likely with several critical humanoid components (Exhibit 3). Overall, components can be categorized as having a low, medium, or high risk of bottlenecks.

Exhibit 3

Supply chain bottlenecks are most likely to emerge around actuators and sensing and perception systems.

Risk of bottlenecks for humanoid robotics systems components

Low Scaled supply already exists in adjacent industries; some adaptations required for humanoid use

Medium Scaled supply already exists in adjacent industries; more extensive adaptations required for humanoid use

High Structural constraints will emerge

Sensing and perception systems:

- Vision systems (LiDAR/depth)
- Encoders (rotary and linear)
- 6-axis force/torque sensors
- Linear force sensors
- Tactile sensors

Battery modules:

- Battery packs
- Thermal/cooling systems

Actuators (linear):

- Ball screws
- Linear actuator integration
- Linear guides
- Planetary roller screws

Structural components:

- Frames and housing
- Precision mechanical parts
- Commodity mechanical

Compute and control platforms¹:

- AI compute /GPU modules
- Real-time control platforms
- Safety controls

Actuators (rotary):

- Driver board/power electronics
- Motor (brushless direct current motor and permanent magnet synchronous motor—neodymium-iron-boron [NdFeB])
- Bearings (cross-roller)
- Encoder
- NdFeB magnets (in motors)

Gearbox

- Planetary gearboxes
- Cycloidal drives
- Harmonic/strain-wave drives



¹System-level bottleneck (integration and safety), not component capacity.

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Low risk: Scaled supply but minor adaptations required

Brushless DC (BLDC) and permanent magnet synchronous motors (PMSM) are produced at scale for EVs, industrial automation, and consumer electronics. Power electronics, standard bearings, camera modules, and LiDAR/radar systems have similarly large and diversified supplier bases. **Battery cells** are supplied by global leaders such as CATL, LG Energy Solution, Panasonic, and Samsung SDI. EV-driven scale has compressed both cost and development timelines for these suppliers. While these components are produced at scale in adjacent industries, they must be integrated and certified for humanoid applications. These adjustments are not trivial, but there is a significant opportunity for suppliers to scale these components, supported by strong existing industrial capacity.

China's distinctive supply chain model

China's humanoid robotics supply chain operates under conditions that are structurally different from those of the rest of the world. These differences are systemic rather than marginal, shaping not only how quickly hardware can scale, but also the underlying cost trajectory and the country's competitive positioning among regional ecosystems.

Manufacturing ecosystem depth

China's humanoid hardware stack benefits from deep adjacency to its electric-vehicle (EV) and broader electromechanical manufacturing base. Many of the most cost- and performance-critical subsystems, including motors, harmonic drives, power electronics, battery systems, and sensors, sit close to EV value chains, enabling supplier reuse, process transfer, and faster scale-up. In addition to accounting for 90 percent of permanent magnet processing globally, China holds strong shares in precision bearings (40 percent), motors (35 percent), and power electronics (30 percent).

At the industrial-robot level, China recorded 295,000 new installations in 2024 and reached an operational stock of approximately 2.03 million robots in factories—both world records—representing roughly 54 percent of global deployments that year.¹ China's extensive electromechanical supplier base enables scaling of humanoid components through reuse of existing industrial products and manufacturing capabilities. In contrast, US and European ecosystems are less tightly coupled to high-volume domestic electromechanical manufacturing. They retain strengths in advanced system design and precision engineering, but supply chain spillover

from adjacent high-volume sectors is structurally weaker.

Cost and speed

China's manufacturing base provides structural cost advantages across actuator-intensive and electromechanical categories, driven by scale economies, component clustering, supplier competition, and reuse of mature EV-grade ecosystems. These are not primarily labor-driven advantages. A recent analysis found that building Tesla's Optimus Gen 2 without Chinese suppliers would cost approximately three times as much, with the bill of materials surging from roughly \$46,000 to \$131,000.²

China's cost advantage shows up in market pricing: Unitree, a Chinese company, publicly lists its G1 robot starting at approximately \$13,500³; costs for Tesla's Optimus are expected to decline to roughly \$20,000 to \$30,000 at scale, depending on configuration and volume assumptions. Dense supplier ecosystems and lower regulatory gating also enable faster hardware iteration. China's robotics ecosystem is geographically concentrated in dense industrial clusters (Hangzhou, Ningbo, Shenzhen, Suzhou), allowing OEMs to source motors, computer numerical control (CNC) housings, printed circuit boards, gear systems, sensors, and battery assemblies within a short radius and to multisource quickly.

Innovation and capital

Innovation indicators skew heavily toward China, with recent data indicating the country filed approximately 7,700 humanoid-related patents over the past

five years, versus roughly 1,560 in the United States and 1,100 in Japan.⁴ China's robotics push is underpinned by industrial policy and capital structures that can support scaling before full commercial validation.

At the same time, constraints remain. According to Stanford University's 2025 AI Index report, US private AI investment reached approximately \$109 billion in 2024, versus roughly \$9.3 billion in China, a roughly 12-fold gap that suggests China's manufacturing and coordination advantages coexist with a structural constraint in private-capital intensity at the frontier AI layer.⁵ Access to advanced compute is further limited by US export controls on certain computing integrated circuits and semiconductor manufacturing equipment. Export markets may also impose limits related to data security, cloud connectivity, and standards, potentially accepting Chinese hardware while restricting software or data flows.

The likely outcome: A bifurcated ecosystem

Rather than a single global winner, the humanoid supply chain is more likely to bifurcate. China may reach hardware scale and early cost compression sooner through manufacturing execution and deployment learning, while US and European ecosystems differentiate via frontier AI sophistication, system architecture, and high-assurance, safety-certified deployments. This bifurcation has direct implications for sourcing strategy, competitive positioning, and the geographic shape of the emerging humanoid economy.

¹ "World Robotics 2025 report—INDUSTRIAL ROBOTS—released by IFR," IFR press release, September 25, 2025.

² "China packs a patent punch in the race to build humanoid robots," *South China Morning Post*, December 20, 2025.

³ UnitreeRobotics, humanoid robot product page, 2026.

⁴ "China packs a patent punch in the race to build humanoid robots," *South China Morning Post*, December 20, 2025.

⁵ 2025 AI Index report, Stanford University Human-Centered Artificial Intelligence, April 2025.

Medium risk: Scaled supply but major adaptations required

This middle tier contains components that are produced at scale in adjacent industries but require more extensive adaptations for use in humanoids than those in the low-risk category. Encoders, certain real-time control electronics, and inertial measurement units (IMUs) are widely available across consumer, industrial, and aerospace grades, yet not all variants provide the latency, drift stability, and fault tolerance required for dynamic bipedal balance. Vision hardware from autonomous-driving programs is often overspecified, bulky, or power-hungry relative to humanoid constraints, requiring significant repackaging. As with low-risk components, the opportunity lies in adapting scaled supply to humanoid requirements: Underlying capacity is sufficient, but greater emphasis is required on qualification and configuration.

High risk: The choke points where supply struggles to meet ambitions

Three component clusters may be particularly subject to bottlenecks, but these same areas represent some of the most attractive opportunities for suppliers to scale capacity and define emerging standards. The first cluster, which contains precision motor components, includes the following:

- ***Harmonic and strain-wave drives.*** The compact, high-torque-density gearboxes used in humanoid joints remain concentrated among a small group of manufacturers, including Harmonic Drive, Nabtesco, and emerging Chinese players such as Leaderdrive. Production of these high-cost components is precision bound and capital intensive, requiring dedicated tooling, metrology infrastructure, and long qualification cycles. Unlike electronics, where capacity can be added rapidly, harmonic (strain-wave) gearboxes are structurally harder to scale. As humanoid volumes rise, demand could outpace the speed at which suppliers can add qualified capacity, creating a significant opportunity for suppliers that can scale precision manufacturing and establish early leadership positions. While alternative gearbox technologies, such as cycloidal drives, benefit from broader industrial use and more scalable manufacturing, harmonic drives remain more exposed because of tighter precision tolerances and a more concentrated supplier base.
- ***Planetary roller screws.*** These components present an even more acute risk of bottlenecks. This segment includes specialized manufacturers such as SKF, as well as select high-precision Asian suppliers. Compared with ball screws, which benefit from broad machine tool and automation demand, roller screws remain a high-load, high-precision niche with a narrow supplier base, long lead times, and limited substitution options. As humanoid OEMs pursue higher payloads and more dynamic motion profiles (shifting toward hybrid rotary and linear actuation architectures), demand for robotics-grade roller screws could exceed existing suppliers' ability to scale—a trend that could make this segment particularly attractive for suppliers that can expand high-precision capacity early.
- ***Robotics-grade linear guides.*** The bottlenecks here arise from subtler constraints. Global players such as Bosch Rexroth, Hiwin, and THK operate at significant industrial scale, providing sufficient overall capacity, but humanoid applications often require compact, high-durability, low-clearance configurations that narrow the effective supplier pool. The primary bottleneck lies in qualifying and ramping robotics-grade subsets, rather than total volume.
- ***Permanent magnets.*** Upstream, high-torque actuators depend on rare-earth permanent magnets, especially neodymium-iron-boron (NdFeB), to achieve required power density. China controls approximately 69 percent of global rare-earth mining and 90 percent of magnet

processing and refining capacity, and recent export licensing changes have already introduced volatility.³ Elon Musk has publicly noted that magnet supply constraints have affected Optimus production.

The second cluster of high-risk components includes those that enable force and tactile sensing:

- **Six-axis force/torque sensors.** These components, supplied by a limited number of robotics-focused vendors, including ATI Industrial Automation and OnRobot, are calibration intensive and require tight metrology and quality control, with limited automation in production. Unlike motors or control electronics, six-axis force/torque sensors benefit little from automotive or consumer spillover. Scaling depends not only on machining capacity but on expanding high-precision calibration infrastructure, a capability that remains concentrated.
- **Linear-force sensors.** These components face similar dynamics: a compact, high-precision niche with limited standardization and shallow supplier depth.
- **Tactile sensors.** Although these components are also prone to bottlenecks, the risks differ. In this category, many leading solutions originate from start-ups or research-driven companies, rather than large-scale industrial suppliers, creating a fragmented, unstandardized landscape with no dominant architecture.

For the third cluster, compute and control, the risk of bottlenecks results from the absence of cohesive, safety-certified, low-latency humanoid platforms, rather than component scarcity. Most OEMs use off-the-shelf AI compute modules (most prominently Nvidia's Jetson series) alongside automotive-grade microcontrollers and industrial servo controllers from vendors such as Elmo, Novanta (Celera Motion), and Synapticon. These are capable components, but none was designed for humanoid-specific coordination and safety integration. The result is a heterogeneous stack: a GPU-based board for perception, distributed joint controllers, and custom middleware for communication and fault handling. There is no standardized, safety-certified "robot ECU" analogous to an automotive engine control unit.

While vertical integration is possible for compute and control, this route is uncommon. One prominent example is Tesla, which has adapted its automotive full self-driving (FSD) compute stack for Optimus and links inference, motion control, and safety logic within a unified system. Few other humanoid OEMs have the scale or semiconductor capability to replicate this model, however.

Rapidly evolving autonomy architectures are compounding the bottlenecks related to compute and control. As AI capabilities advance, OEMs retain tight control over firmware and system integration, making it hard for external suppliers to design against a stable specification. OEMs shoulder most of the burden for real-time coordination across 20 to 40 primary structural actuators, often at kilohertz-level control frequencies, combined with fail-operational safety logic. This makes compute a systems bottleneck (integration and certification) rather than a component supply bottleneck (capacity constraints), a distinction that matters when prioritizing investment.

³ "Rare earth magnets 2026–2036: Technologies, supply, markets, forecasts," IDTechEx, 2026.

From vertical integration to supplier platform ecosystems

The uneven readiness of the humanoid supply chain directly shapes how OEMs source today. In some cases, OEMs now rely on vertical integration because no viable supplier options exist, not because they have a strategic preference for keeping manufacturing in-house. As the humanoid industry scales, sourcing patterns may change.

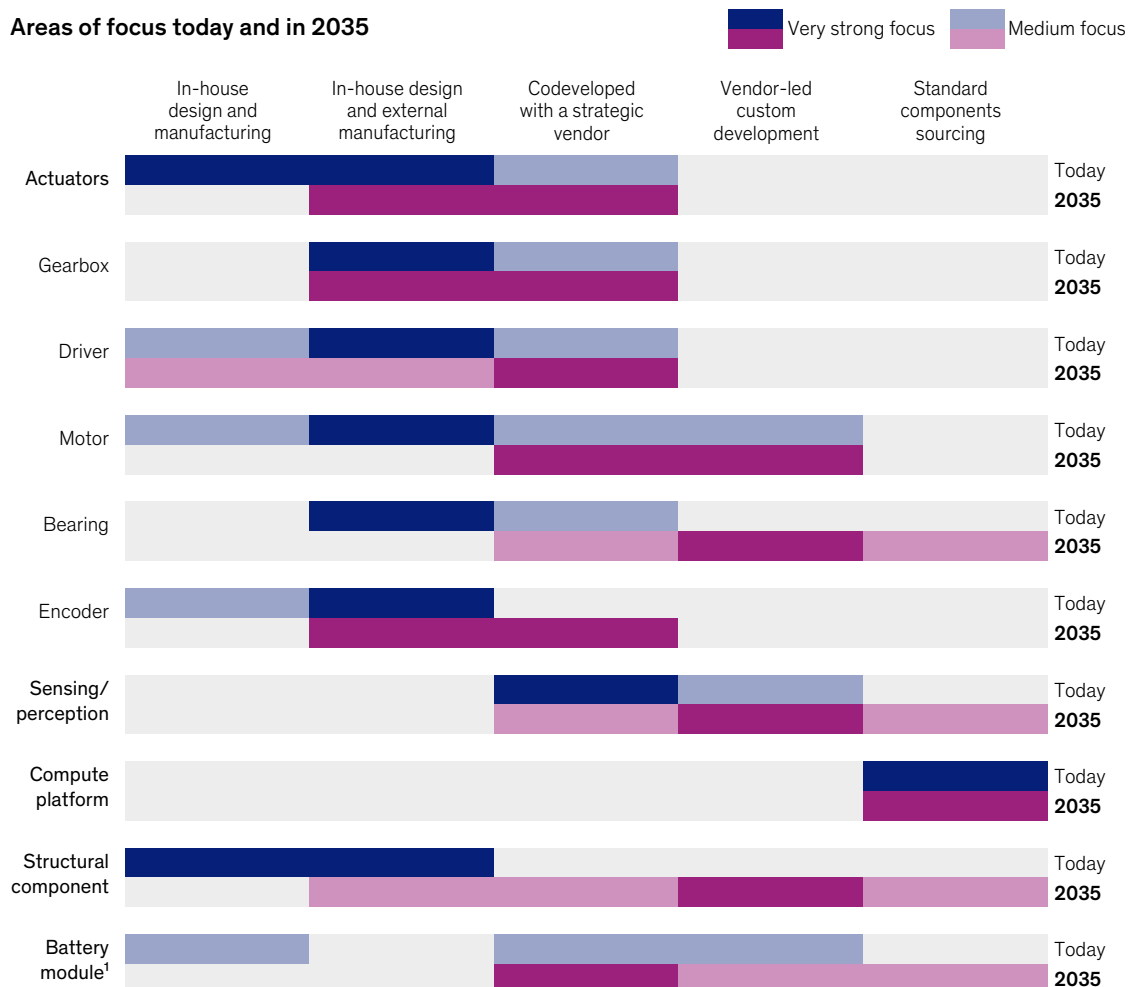
Today's market: OEMs build because they must

Most OEMs now redesign or tightly integrate the subsystems where performance sensitivity and supplier scarcity intersect, especially for actuation (Exhibit 4). As the largest single cost driver and primary performance differentiator, actuators concentrate the core competitive levers: torque density, backdrivability, precision, and lifetime performance (see sidebar "Inside the

Exhibit 4

As the supply chain matures and scale is achieved, OEMs are likely to focus their value add on differentiating domains.

Areas of focus today and in 2035



¹An exemption is likely to apply to humanoid robotics OEMs with existing battery manufacturing sites from electric-vehicle businesses.
Source: Expert interviews, public filings, web search

actuator”). Modern humanoids require dozens of actuators that must be lightweight yet powerful, efficient yet responsive. Tesla’s Optimus, for example, has 28 joint actuators in its body, plus 50 additional actuators in its Gen 3 hands.

There is no equivalent of an “engine supplier” for humanoid actuators. Instead, the industry remains in a premodular phase, with each leading project resorting to custom development. Tesla uses custom-designed frameless motors and gearboxes in its actuators; Agility Robotics initially leveraged off-the-shelf motors for Digit but had to heavily modify and integrate them into proprietary leg modules.

This vertical integration for actuation is a rational response to current conditions. It secures cost performance control, mitigates supplier concentration risk, protects emerging intellectual property (IP), and accelerates iteration. Humanoid OEMs, which have bold aspirations but still face significant tech and timing risks that need to be mastered, cannot provide the long-term volume guarantees required to trigger large-scale supplier R&D and tooling commitments. Hence, vertical integration for actuators is likely to persist until volumes become more predictable. Only then will supplier-led modularization become economically viable at scale.

Inside the actuator

Actuators define a humanoid robot’s core physical capabilities: how strong it is, how fast it moves, how precisely it can control force. A typical integrated actuator consists of five core elements:

- **Gearbox (30 to 50 percent of actuator cost).** Harmonic/strain-wave drives are most widely adopted for humanoid joints due to their compactness and high ratio, but they are prone to failure under backdriving forces (for instance, when a humanoid falls). Cycloidal drives are preferred for large joints requiring high torque and reliability, though at higher cost. Planetary and belt/cable/tendon drives are commonly found in lower-cost humanoids and tend to offer limited lifespan. For linear actuation paths, the analogous choke points are roller screws and high-durability linear guides, where capacity and qualification cycles can be even more restrictive than in rotary reducers.

- **Driver (15 to 20 percent).** This category includes motor control electronics, which are increasingly safety rated.

- **Motor (10 to 20 percent).** Frameless brushless DC motors dominate this category because of their efficiency and availability; permanent magnet synchronous motors offer smoother commutation at higher cost. Coreless motors are used in humanoid fingers. Differentiation typically occurs through custom winding, torque density optimization, and thermal management rather than through fundamental architecture.

- **Mechanical components (10 to 20 percent).** These include structural casing, shafts, and precision bearings (for example, cross-roller and angular contact).

- **Sensors (5 to 10 percent).** Categories include encoders (position), force/torque sensors, and temperature sensors.

While often treated as a single module, differentiation concentrates unevenly across these elements of the actuator stack. Most performance differences result from gearboxes, which determine torque density, compactness, backlash, and durability—factors that directly shape how each joint performs.

Motors, by contrast, are more commoditized, and only a few precision motor suppliers exist. Companies such as Maxon (Switzerland) and Kollmorgen (United States) supply high-end motors for robotics, but demand is outpacing what these mostly industrial-focused companies can deliver. As OEMs pursue higher payloads and faster, safer dynamics, many architectures are shifting from purely rotary joints toward hybrid rotary and linear actuation, increasing exposure to precision linear-motion components.

Advantage will concentrate less in supplying individual components and more in defining integrated subsystem platforms.

By contrast, consider compute platforms and selected sensing modules, which already sit further toward the “buy” side of the sourcing spectrum. These subsystems benefit from established adjacent ecosystems and standardized architectures, making external sourcing economically rational even at today’s scale.

By 2035: Selective modularization, subsystem by subsystem

Over the next decade, the transition from in-house design toward codevelopment and vendor-led modules will not happen uniformly. Components with strong industrial spillovers or scalable manufacturing economics, particularly structural parts, battery systems, and compute/control platforms, are likely to shift toward external sourcing earlier, as safety architectures mature and interfaces standardize. Once humanoid architecture converges, structural components and batteries could become one of the fastest areas to modularize and industrialize (see sidebar “Opportunities in structural components and batteries”).

By contrast, the shift from in-house development for high-performance actuation and robotics-grade force sensing will occur more slowly. These areas combine supplier concentration, performance sensitivity, and IP leverage. Until capacity expands and interfaces stabilize, OEMs will be reluctant to relinquish architectural control. By 2035, a layered structure is likely to emerge: Tier 1 suppliers in compute and battery systems (similar to the automotive supply chain), selective consolidation in sensing, and continued OEM influence over motion architecture.

Overall, three conditions will trigger the broader shift toward industrialization: predictable multiyear volumes, stable subsystem architectures, and converging interface standards. None is fully present today. The inflection point is economic: When predictable multiyear volumes justify dedicated tooling, certification investment, and platform-level R&D, the ability to define standardized interfaces and subsystem form factors will determine who captures long-term ecosystem leverage.

That inflection point may arrive sooner than expected. Even as most OEMs remain vertically integrated, a wave of strategic partnerships over the past 18 months signals that adjacent industries are not waiting for the premodular phase to end. They are positioning now for the platform roles that will eventually define the ecosystem. This points to a broader shift in how value will be captured across the ecosystem: Advantage will concentrate less in supplying individual components and more in defining integrated subsystem platforms, such as actuator modules or compute and control stacks.

Opportunities in structural components and batteries

Beyond actuators, sensing, and compute, suppliers may find meaningful opportunities in two additional hardware domains.

Structural components

Structural parts (frames, joint housings, brackets, enclosures) account for roughly 5 to 10 percent of the humanoid bill of materials (BOM) and provide the physical backbone of the robot. Today's humanoids are still largely built using prototype-grade methods: computerized numerical control (CNC) machining, low-volume fabrication, and custom assembly. Designs iterate rapidly, and geometry is not yet fully frozen, limiting the feasibility of high-volume tooling investments.

For structural components, the constraint is not technological scarcity but design maturity and volume uncertainty. Until architecture stabilizes, suppliers have limited incentive to invest in robotics-specific production capacity. As a result, structural manufacturing remains fragmented with limited standardization across OEMs.

Once designs converge, however, structural components could become one of the fastest areas to modularize and industrial-

ize, representing one of the clearest pathways toward cost compression. Automotive and industrial manufacturing suppliers, particularly those experienced in die casting, forging, and lightweight composite integration, are well positioned to transition humanoid structures from CNC-based prototyping to scalable production. Suppliers that support early codesign with OEMs, helping freeze geometries and optimize for tooling, may anchor themselves in future high-volume platforms.

Battery modules

Battery systems enable untethered operation and account for roughly 5 to 10 percent of the humanoid BOM. Most OEMs now procure standard cylindrical or pouch cells from established electric-vehicle (EV) suppliers and then design the pack architecture in-house. Battery cell supply is not likely to be constrained by 2035; the differentiator is pack architecture and operational uptime strategy.

Humanoids require high levels of burst power for dynamic motion, stable thermal behavior in compact enclosures, and robust protection in case of falls or colli-

sions. These requirements differ meaningfully from EV battery modules, which are optimized for sustained discharge and range. Battery suppliers are beginning to address this gap directly:

- Samsung SDI unveiled solid-state batteries designed specifically for humanoid robots at InterBattery 2026, targeting mass production in the second half of 2027.¹
- Hyundai Motor Company and Samsung SDI signed a memorandum of understanding in February 2025 to codevelop robot-dedicated batteries.²
- LG Energy Solution has been confirmed as the battery supplier for Boston Dynamics' Atlas, using 46-series cylindrical nickel-cobalt-manganese cells for a humanoid scheduled to deploy in Hyundai's US factories starting in 2028.³

These moves suggest the battery domain is developing humanoid-specific product lines faster than previously expected.

¹ Sarah, Chea, "Samsung SDI to unveil batteries for humanoid robots at this week's InterBattery 2026 event," *Korea JoongAng Daily*, March 9, 2026.

² Kan Hyeon-woo, "Collaboration of industry giants: Hyundai, Samsung team up on batteries for robots," *Korea Herald*, February 25, 2025.

³ LG Energy Solution confirmed as battery supplier for Boston Dynamics' Atlas humanoid.

In actuation, automotive and precision-motion suppliers are staking claims to the "engine maker" role that does not yet formally exist. A large automotive and industrials supplier signed three humanoid actuator partnerships in five months (Neura Robotics,⁴ UK-based Humanoid, and Chinese manufacturer Leju Robotics⁵), becoming the preferred actuator supplier across wheeled and bipedal platforms and codeveloping next-generation strain-wave gear actuators. The company expects up to 10 percent of group sales in 2035 from new sectors, including humanoid

⁴ Amir Orusov, "Schaeffler partners with Neura Robotics to develop and supply humanoids," *Reuters*, November 4, 2025.

⁵ "Humanoid robotics: Schaeffler expands global partner network and cooperates with Leju Robotics in China," Schaeffler press release, March 2, 2026.

robotics. Bosch has entered through a partnership with Neura Robotics covering component supply and motor production,⁶ while its Boyuan Capital arm formed a joint venture with Chinese humanoid developer Galbot.⁷ Magna took an equity stake in Sanctuary AI and is applying its automotive manufacturing capabilities to humanoid scalability.⁸ The pattern is consistent: These companies are codeveloping humanoid-specific subsystems rather than adapting off-the-shelf parts.

In compute, a parallel land grab is underway. Qualcomm, for example, launched the Dragonwing IQ10, a humanoid-specific processor, and is working with players such as Figure AI and Neura Robotics to define next-generation compute architectures as humanoid platforms scale.⁹ Recently, Infineon, NXP Semiconductors, STMicroelectronics, and Texas Instruments all formalized humanoid-specific product integrations alongside Nvidia, covering motor control, real-time communications, sensor fusion, and safety logic. These are products ready for shipping, not research projects.

In manufacturing, Jabil has become the worldwide production partner for Appttronik's Apollo humanoids, illustrating how contract manufacturers are positioning themselves as scalable production platforms.¹⁰ SoftBank announced its \$5.4 billion acquisition of ABB's robotics division, emphasizing that the company's "next frontier is physical AI." This move builds on a trend of investments in robotics—such as Midea's acquisition of Kuka and Amazon's targeted purchases of companies such as Fauna Robotics, Rightbot Technologies, and Rivr—and signals investor recognition that supply chain access will significantly shape the physical-AI landscape.¹¹

The most open platform opportunity remains within sensing: No widely adopted humanoid sensor suite exists, and OEMs still assemble and calibrate disparate cameras, depth sensors, IMUs, and force/torque sensors themselves.

The actions that all stakeholders take over the next few years will determine whether the humanoid supply chain matures fast enough to scale.

⁶ "Neura partners with Bosch to advance humanoid robotics," Evertiq, January 23, 2026.

⁷ "Boyuan Capital (investment platform under Bosch Group) and Galbot forged JV," GlobeNewswire, June 23, 2025.

⁸ Thomas Seal, "Robotics startup Sanctuary signs deal for factory tests, funds," Bloomberg, April 11, 2024.

⁹ "Qualcomm introduces a full suite of robotics technologies, powering physical AI from household robots up to full-size humanoids," Business Wire, January 5, 2026; "NEURA Robotics and Qualcomm enter strategic collaboration to advance physical AI and cognitive robotics," Qualcomm press release, March 9, 2026.

¹⁰ "Appttronik and Jabil collaborate to scale production of Apollo humanoid robots and deploy in manufacturing operations," Jabil press release, February 25, 2025.

¹¹ John Revill and Anton Bridge, "SoftBank to buy ABB's robot business for \$5.4 billion in push to merge AI and robotics," Reuters, October 8, 2025.

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Strategic imperatives: What suppliers should do now

For suppliers, the window to shape the humanoid robotics supply chain is open but narrowing. Design successes at the prototype stage convert into production incumbency once architectures stabilize and volumes scale. To understand how those that move fast may be able to claim positions quickly, consider the trajectory for one large automotive and industrials supplier, which went from having no humanoid presence to becoming a key actuator supplier in under two years.

As suppliers plan their next move, five priorities stand out:

- ***Forging early codevelopment partnerships.*** Suppliers could gain an advantage by working with OEMs to embed their components into humanoids before specifications are finalized. OEMs are actively seeking such external partners and resort to vertical integration out of necessity, not preference.
- ***Investing in safety and certification.*** Humanoids operating alongside people must meet stringent standards. Suppliers that proactively build functional safety capabilities, such as SIL/ASIL certification, redundant design capabilities, or compliance with ISO robotics criteria, will lower adoption friction and become preferred choices, particularly in Western markets.
- ***Standardizing and designing for modularity.*** Suppliers could build interoperable components that serve multiple robot platforms and help define emerging interface standards. Standardization will broaden addressable demand and increase switching costs, thereby benefitting early movers.
- ***Proving scalable manufacturing.*** OEMs will prioritize suppliers that demonstrate credible volume readiness: automation, secured critical inputs (including rare-earth supply), and consistent quality control. Partnerships with automotive or EMS firms can accelerate ramp capabilities.
- ***Building life cycle revenue streams.*** Spare parts, maintenance contracts, upgrade kits, and predictive diagnostics will become material profit pools. In uptime-sensitive applications, service capability may be as decisive as hardware performance.

The humanoid value chain is forming now, and its trajectory will be shaped not by any single player but by coordinated movement across the ecosystem. OEMs will need to stabilize architectures and signal volumes credibly enough to trigger supplier investment; suppliers will need to invest ahead in modular, safety-certified capabilities to be prepared for the surge in demand; and end users can engage in real-world deployments that validate the business case and generate the operational data humanoids need to improve. None of these shifts can happen in isolation; they require close coordination along the value chain. The actions that all stakeholders take over the next few years will determine whether the humanoid supply chain matures fast enough to scale, and how quickly cost, complexity, and fragmentation can be overcome to unlock deployment.

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